MIL-I-12822C(MU)
10 February 1969
SUPERSEDING
NIL- I-12882B (Mu)
18 JUNE 1965

MILITARY SPECIFICATION

IGNITER, ROCKET MOTOR, M26 PARTS AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers tthe parts and loading, assembling and packing for one type of igniter designated as Igniter, M26, for use in Rocket Motor, M4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal from a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-I045607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-414	- Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single anti Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1340

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DRAWINGS

ARMY

7548439 - Igniter, M25, Loading Assembly
7548839 - Carton, Packing, Ammunition for Igniter, Rocket Motor, M26, Assembly, Details and Packing
7548840 - Box, Packing, Ammunition for Igniter, Rocket Motor, M26, Assembly, Details and Packing

PUBLICATIONS

ARMY

ET-7548439 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Igniter. - The igniter shall comply with all requirements specified on Drawing (dwg.) 7548439, and with all requirements specified in applicable specifications.

3.3 Black powder.

3.3.1 Moisture content. - The moisture content of the black powder at the loading plant, at the time of loading, shall not exceed 0.5 percent.

3.3.2 Exposure to air. - The black powder, after determination of the misture content, shall not be exposed to air during weighing and loading operations more than ten minutes total accumulated time.

3.4 Functioning.

3.4.1 Time. - The igniters shall function in not less than 14 seconds and not more than 16 seconds when tested as specified in 4.4.4.

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3.4.2 Duds. - The ignitor shall fire when tested as specified in 4.4.4.

Workmanship. - The item of shipment, its componenets and materials shall be free of dirt, chips, grease, corrosion, anti other foreign material. There shall be no cracks, scratches, burrs, dents, gouges, holes, porous areas, distortions, or other undesirable characteristics which would, in any way, detract from the usability of the product for its intended purpose. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The The c paintings and coatings on all components shall be restored when such paintings anti coatings have been marred or obliterated.

3.6 First Article Inspection. - This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract .

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all ispeciton requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the preformance of the inspection requirements specified herein, unless (disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements .

4.1.1 Submission lof product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification] number and date-, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance. are on file and available for review.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying orgtanization. The initial certificate submitted shall be substantiated by evidence of the agent’s authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) loaded igniters and ten (10) complete sets of parts (I.e., ten (10) of each individual component and ten (10) of each subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that It conforms to the requirements of the contract and submit a record of findings. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period In excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Assemblies, subassemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any assembly, subassembly or component falls to comply with any of the applicable require-ments, the. First Article Sample shall be rejected. The Govenrment reserves the right to terminate its inspection upon any failure of an assembly, subassembly or component to comply

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4.3 Inspection Provisions

4.3.1 Lot Formation. - The tern "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogenous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by on manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety , performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 or MIL-STD-414 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the Contracting Officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the item of delivery, shal be homogeneous and of a size convenient to the contractor and the Government Inspector.

4.3.2 Examination. - Sampling plans and procedures for for the following classifications of defects shall be in accordance with Standard MIL-STD-105 or MIL-STD-414 except that inspection for critical defects shall be 100 percent. continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0. 40 percent for

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each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessay for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Retainer relay (dwg. 7548452)

Categories	Defects	Method, of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 percent		
201	Length.....	Gage	01001
202	Outside diameter.....	Gage	01002
203	Split or crack in aluminum.....	Visual	01003
205	Shellac missing.....	Visual	01005
206	Evidence of poor workmanship..	Visual	01006

4.3.2.2 Retainer relay assembly (dwg. 7543451)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Disc improperly assembled....	Visual	02001
202.	Disc perforation missing.....	Visual	02002
203.	Shellac on outer surface of disc.....	Visual	02003

4.3.2.3 Delay Element Assembly (dwg. 9209521)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 100 percent		
101.	Primer paste missing.....	Visual	03001
Minor:	None defined.		

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4.3.2.4 Delay assembly, prior to crimping (dwg. 7548442)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
	101. Weight of charge, min.	Balance	04001
	102. Charge missing.	Visual	04002
Minor:	None defined.		

4.3.2.5 Igniter, loading assembly, prior to assembly of sleeve (dwg. 7503439)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
	101. Igniter charge weight incorrect.....	Balance	05001
	102. Component or operation missing.....	Visual	05002
Minor:	None defined.		

4.3.2.6 Ignitor loaded assembly (dwg. 7548439)

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Explosive escaping.....	Visual	06001
Major:	None defined.		
Minor:	AQL 1.50 percent		
	201. Total length maximum.....	Gage	06002
	202. Heat seal incomplete.....	Visual	06003
	203. Foreign matter.....	Visual	06004
	204. Location of closure.....	Visual	06005

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4.3.2.7 Cup assembly prior to assembly into delay assembly (dwg. 7548445)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Closing disc missing or improperly assembled.....	Visual	07001
202.	Cup scratched or incompletely coated.....	Visual	07002
203.	Shellac on *closing* disc.....	Visual	07003

4.3.2.8 Closure (dwg. 7548450)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside diameter.....	Gage	08001
102.	Large outside diameter.....	Gage	08002
103.	Angle of outside surface improper.....	Gage	08003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship.	Visual	08004

4.3.2.9 Sealed barrier bag (dwg. 7548839)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
102.	Waterproof seal broken by puncture, tear or cut.....	Visual	09001
Minor:	AQL 1.00 percent		
201.	Markings missing, illegible, or incorrect.....	Visual	09002
202.	Contents loose.....	Manual	09003

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4.3.2.10 Sealed wood packing box (dwg. 7548840)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken or loose	Visual/Manual	10001
102.	Board broken	Visual	10002
103.	Marking missing, illegible, or incorrect	Visual	10003
Minor:	AQL 1.00 percent		
201.	Strapping improperly assembled	Visual/Manual	10004
202.	Contents loose	Manual	10005

4.3.3 Testing

4.3.3.1 First Article sample (see 4.2)

4.3.3.1.1 Inert parts. - The sample submitted as inert parts and subassemblies shall be inspected by the Government for conformance to all drawing requirements.

4.3.3.1.2 Loaded assemblies. - The samples submitted as loaded assemblies shall be selected as specified below after having been examined for any or all other drawing requirements.

4.3.3.1.2.1 Moisture content of black powder (see 3.3.1) - Code No. 11001. Two (2) igniters shall be sectioned and tested as specified in 4.4.1. The black powder moisture content of the loaded igniters shall comply with the requirement specified In MIL-P-223, Class 7.

4.3.3.1.2.2 Pull test of delay assemblies (see dwg. 7548442) - Code No. 12001. Three (3) igniters shall be tested as specified in 4.4.2.

4.3.3.1.2.3 Vacuum test (see dwg. 7548439) - Code No. 06006. This test shall be conducted 100 percent as specified in 4.4.3.

4.3.3.1.2.4 Functioning test (see 3.4) - Code No. 06007. This test shall be conducted 100 percent as specified in 4.4.4.

4.3.3.2 Production testing.

4.3.3.2.1 Determination of moisture content of black powder (See 3.3.1) - Major defect - Code No. 11001 - The contractor shall provide adequate controls to assure the compliance of each lot with the specified requirement. For verification, the

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contractor shall select at least one (1) random sample of approximately one ounce of black powder from the container of powder in use for each eight hours of production or fraction thereof that the line operates. Failure to comply with the applicable requirement, shall be cause for rejection of the eight hours production represented by the sample. The test shall be performed as specified in 4.4.1.

4.3.3.2.2 Pull test of delay assemblies (see dwg. 7548442) - Major defect - Code No. 12001. - The pull test shall be conducted 100 percent in the manner specified in 4.4.2. Damage to or extension of the parts shall be cause for rejection of the assembly.

4.3.3.2.3. Vacuum test (see dwg. 7548439) - Major defect - Code No. 06006. - The vacuum test shall be conducted 100 percent in the manner specified in 4.4.3. Any assembly exhibiting leakage shall be rejected and removed from the lot.

4.3.3.2.4 Functioning test (see 3.4) - Major defect - Code No. 06007.

4.3.3.2.4.1 At 120 degrees F. - The sampling plan for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-1, Code letter K with AQL of 0.65 percent for each limit separately. Duels shall be replaced with additional igniters. The test shall be performed as specified in 4.4.4.

4.3.3.2.4.2 At 0 degrees F. - The sampling plan for this test shall be as specified in 4.3.3.2.4.l.

4.3.3.2.4.3 Duds. - If no duds are discovered during the performance of either, or both, phases of the test, the lot may be accepted. If one or two duds are discovered in either phase, a second sample of 70 igniters shall be selected, to test for duds only., under the conditions of the particular phase in which the duds occurred. If at any time during the test, or retest, a total of 3 duds occur, the lot shall be rejected without further test.

4.3.4 Inspecition Equipment - Equipment Tabulation Number ET-7548439 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the-requirements of the ET may be used by DOD activities

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for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever Inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content of black powder. - The moisture content of black powder shall be determined as specified in specification MIL-P-223.

4.4.2 Pull test of delay assemblies. - The delay assemblies shall be subjected to the pull test specified on dwg. 7548442 in the manner described thereon. Test equipent shall be in accordance with 4.3.4.

4.4.3 Vacuum test. - The igniter assemblies shall be subjected to the vacuum specified on dwg. 7548439 In the manner described thereon. The test shall be conducted using testing equipment as specified in 4.3.4.

4.4.4 Functioning

4.4.4.1 One hundred twenty degrees Fahrenheit (120 F) phase. - Sample igniter assemblies selected as specified In 4.3.3.2.4 shall be conditioned at 120 degrees F plus or minus 2 degrees F for a period of not less than 2 hours and tested within 2 minutes after removal from the conditioning chamber. The assembly shall be ignited with a M60 blasting fuze Igniter. Observation shall be made for compliance with 3.4. Functioning time shall be recorded to the nearest 0.1 second. Test Test equipment shall be in accordance with 4.3.4.

4.4.4.2 Zero degrees F phase. - This test shall be performed as specified in 4.4.4.1 except that the conditioning temperature shall be O degrees plus or minus 2 degrees F.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level B. - The igniters shall be packaged In a carton in accordance with dwg. 7548839.

5.2 Packing

5.2.1 Level B, - The igniters shall be packed in a wood box In accordance with dwg. 7548840.

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6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot In accordance with the information specified In Standard MIL-STD-1167.

6.2 Inspeciton code numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspecion equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of Inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings In accordance with 6.3.4 for all the equipment coded as “Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply-with, and use.

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these designs accordingly. The contractor may, however, In connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted In accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other desibns of inspeciton equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of Inspection equipment designs is permissible and encouraged. However, the Arsenal cpompletion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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